

A Nonlinear Degenerate Advection-Diffusion Model for Crowd Instability and Stampede Risk Prediction

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Abstract

Crowd disasters arise from nonlinear interactions between density, spatial constraints, and limited mobility. We propose a nonlinear degenerate advection-diffusion system for pedestrian density $\rho(\mathbf{x}, t)$ that captures three behavioral regimes: potential-driven flow at low density, diffusive spreading at moderate density, and nonlinear compression at high density. The model combines a harmonic potential field, density-dependent velocity, weak diffusion, and a smooth super-quadratic pressure law. A risk functional $\mathcal{R}(\mathbf{x}, t)$ is introduced to identify high-risk regions through rapid density growth, reduced mobility, and proximity to geometric constraints. The framework provides a computationally efficient alternative to agent-based simulations while retaining key instability mechanisms relevant to crowd congestion and evacuation scenarios.

1 Introduction

Large-scale crowd motion exhibits complex nonlinear behavior that cannot be captured by linear or purely diffusive models. In confined environments, the interplay between crowd density, geometry, and exit structure can lead to congestion, stop-and-go flow, and localized instability. Events such as the Mina stampede (2015) and the Love Parade disaster (2010) demonstrate that local density increases can propagate and amplify, creating hazardous conditions.

Existing approaches face several limitations:

- Microscopic models scale poorly with population size.
- Classical macroscopic models often assume smooth flux functions and do not explicitly represent density saturation.
- Kinetic approaches may neglect strong spatial congestion effects and the influence of geometry on flow redistribution.

The central challenge is modeling the transition from smooth crowd flow to congested, unstable dynamics. This work introduces a nonlinear continuum framework that incorporates density-dependent mobility and pressure-driven effects to capture this transition in a mathematically tractable way.

2 Mathematical Model

2.1 Harmonic Potential Field

The environment is encoded through a harmonic potential field:

$$\nabla^2 \phi = 0 \quad \text{in } \Omega, \quad (1)$$

$$\phi = 0 \quad \text{on } \partial\Omega_{\text{exit}}, \quad (2)$$

$$\nabla \phi \cdot \mathbf{n} = 0 \quad \text{on } \partial\Omega_{\text{wall}}. \quad (3)$$

The gradient $-\nabla \phi$ provides a direction field that guides the movement toward exits. This formulation follows continuum pedestrian flow models [1].

2.2 Density-Dependent Velocity

Pedestrian velocity decreases with density:

$$\mathbf{v}[\rho] = -\nabla \phi \left(1 - \frac{\rho}{\rho_{\text{max}}}\right)^\beta, \quad \beta > 0. \quad (4)$$

This ensures vanishing mobility as $\rho \rightarrow \rho_{\text{max}}$, reflecting the reduced freedom of movement in crowded conditions.

2.3 Density Evolution Equation

The density evolves according to:

$$\partial_t \rho + \nabla \cdot (\rho \mathbf{v}) = D \nabla^2 \rho - \nabla \cdot (\rho \nabla P(\rho)). \quad (5)$$

The terms represent:

- Advection: directed motion toward exits
- Diffusion: small-scale randomness and smoothing
- Nonlinear pressure: crowd compression effects

This is a nonlinear advection-diffusion system with density-dependent flux and degeneracy at high density.

2.4 Smooth Pressure Law

To avoid discontinuities, we define:

$$P(\rho) = k \left(\frac{\rho}{\rho_{\text{crit}}}\right)^n S(\rho), \quad n > 2, \quad (6)$$

where $S(\rho)$ is a smooth activation function, for example:

$$S(\rho) = \frac{1}{1 + e^{-a(\rho - \rho_{\text{crit}})}}.$$

This ensures:

- negligible pressure at low density
- rapid growth near ρ_{crit}
- differentiability for numerical stability

2.5 Risk Functional

We define a risk indicator:

$$\mathcal{R}(\mathbf{x}, t) = \alpha \frac{\rho}{\rho_{\text{max}}} + \delta \frac{1}{d(\mathbf{x}) + \epsilon} + \gamma \frac{1}{|\mathbf{v}| + \epsilon} + \eta \Psi(\rho) - c(\mathbf{x}, t), \quad (7)$$

where

$$\Psi(\rho) = \begin{cases} 0, & \rho < \rho_{\text{crit}}, \\ \left(\frac{\rho - \rho_{\text{crit}}}{\rho_{\text{max}}} \right)^2, & \rho \geq \rho_{\text{crit}}. \end{cases}$$

This functional increases with density, proximity to exits or constrictions, and reduced mobility. The term $d(\mathbf{x})$ denotes the local distance to the nearest exit or geometric restriction.

2.6 Camaraderie Term

Cooperative effects are modeled as:

$$c(\mathbf{x}, t) = \frac{G(\mathbf{x}, t)}{N_{\text{local}}} (1 - I(\mathbf{x}, t)) \left(1 - \frac{\rho}{\rho_{\text{max}}} \right)^m, \quad m > 1. \quad (8)$$

This term decreases with increasing density, reflecting reduced coordination in crowded conditions.

2.7 Model Parameters

The model contains several physically interpretable parameters governing mobility, congestion, and risk estimation. Representative values used in the simulations are summarized in Table 1.

Parameter	Description	Value
ρ_{max}	Maximum pedestrian density	8 persons/m ²
ρ_{crit}	Critical congestion density	4 persons/m ²
D	Diffusion coefficient	0.02
β	Mobility exponent	2
k	Pressure coefficient	1.0
n	Pressure exponent	3
a	Pressure activation steepness	5
ϵ	Regularization constant	10 ⁻³

Table 1: Representative parameters used in numerical experiments. These values are intended to demonstrate qualitative crowd behavior and are not calibrated to a specific dataset.

3 Theoretical Properties

3.1 Mass Conservation

Integrating the governing equation over the computational domain Ω gives

$$\frac{d}{dt} \int_{\Omega} \rho d\Omega = - \int_{\partial\Omega} F \cdot \mathbf{n} dS,$$

where F denotes the total pedestrian flux.

Therefore the total population remains conserved except through designated exits. No pedestrians are artificially created or destroyed within the domain.

3.2 Positivity Preservation

Provided that

$$\rho(\mathbf{x}, 0) \geq 0,$$

the advection-diffusion-pressure system preserves nonnegative density values throughout the simulation.

This property ensures physical consistency and prevents the appearance of nonphysical negative densities.

3.3 Congestion Regulation

As pedestrian density approaches the maximum allowable density,

$$\rho \rightarrow \rho_{\max},$$

the mobility term

$$\left(1 - \frac{\rho}{\rho_{\max}}\right)^{\beta}$$

approaches zero.

Consequently pedestrian motion naturally slows in highly congested regions, preventing unrealistic acceleration and improving numerical stability.

4 Regime Behavior

The system exhibits distinct regimes:

- Low density: advection-dominated flow toward exits
- Moderate density: diffusion smooths fluctuations
- High density: nonlinear pressure induces strong gradients and congestion

Rather than true shock formation, the model produces steep gradients and localized clustering consistent with observed crowd behavior.

5 Numerical Methodology

5.1 Computational Domain

Simulations were performed on two-dimensional domains representing pedestrian facilities containing walls, corridors, bottlenecks, and exits.

Walls were modeled using reflective no-flux boundary conditions, while exits were represented as absorbing boundaries through the harmonic potential formulation.

Initial pedestrian density was assigned to designated holding regions and evolved according to the nonlinear governing equations.

5.2 Numerical Scheme

The governing equations were solved using a finite-volume discretization.

Advection terms were evaluated using an upwind flux approximation to preserve stability near steep density gradients. Diffusive and pressure-driven terms were discretized using second-order finite differences.

Time integration employed a semi-implicit scheme, allowing stable simulation of high-density congestion while maintaining computational efficiency.

5.3 Simulation Scenarios

Two representative crowd configurations were investigated:

- Multi-branch bottleneck evacuation
- Stadium evacuation with a narrow exit corridor

These scenarios were selected to examine density accumulation, congestion propagation, and the formation of high-risk regions under realistic geometric constraints.

6 Numerical Results

To evaluate the proposed framework, a series of crowd-flow simulations were performed under varying geometric constraints and population distributions. The objective was to investigate the emergence of congestion, bottleneck formation, and high-risk regions under realistic evacuation conditions.

The simulations were conducted on a two-dimensional domain representing pedestrian facilities with exits, corridors, and obstacle boundaries. Density evolution was computed using the nonlinear advection-diffusion formulation introduced above, while crowd movement was guided by the harmonic potential field.

6.1 Multi-Branch Bottleneck Scenario

Figure 1 shows the evolution of density in a symmetric multi-branch evacuation environment. Four crowd reservoirs are connected to a central corridor through narrow connecting passages.

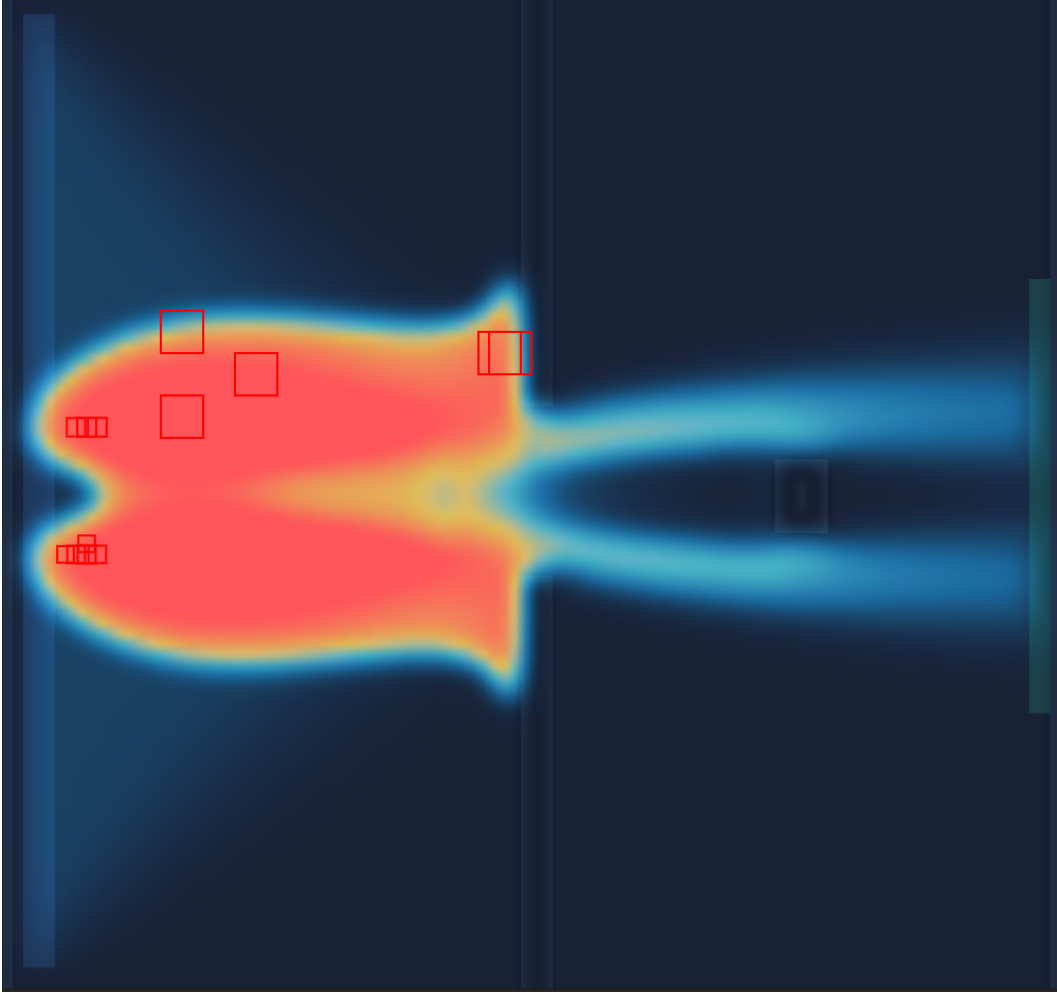


Figure 1: Density distribution in a multi-branch evacuation scenario. High-density regions develop within the holding areas before pedestrians merge into the central corridor. The harmonic potential field naturally guides flow toward the common exit route while the non-linear pressure term limits excessive crowd compression.

The simulation demonstrates several notable features:

- Density accumulates near the entrances of the connecting passages.
- Flow remains organized despite the interaction of multiple incoming streams.
- Congestion remains localized around geometric restrictions.
- High-density regions are successfully identified by the risk functional.

The emergence of localized density peaks confirms the model's ability to detect potential instability zones before complete congestion develops.

6.2 Stadium Rush Scenario

Figure 2 presents a larger-scale evacuation toward a narrow exit corridor.

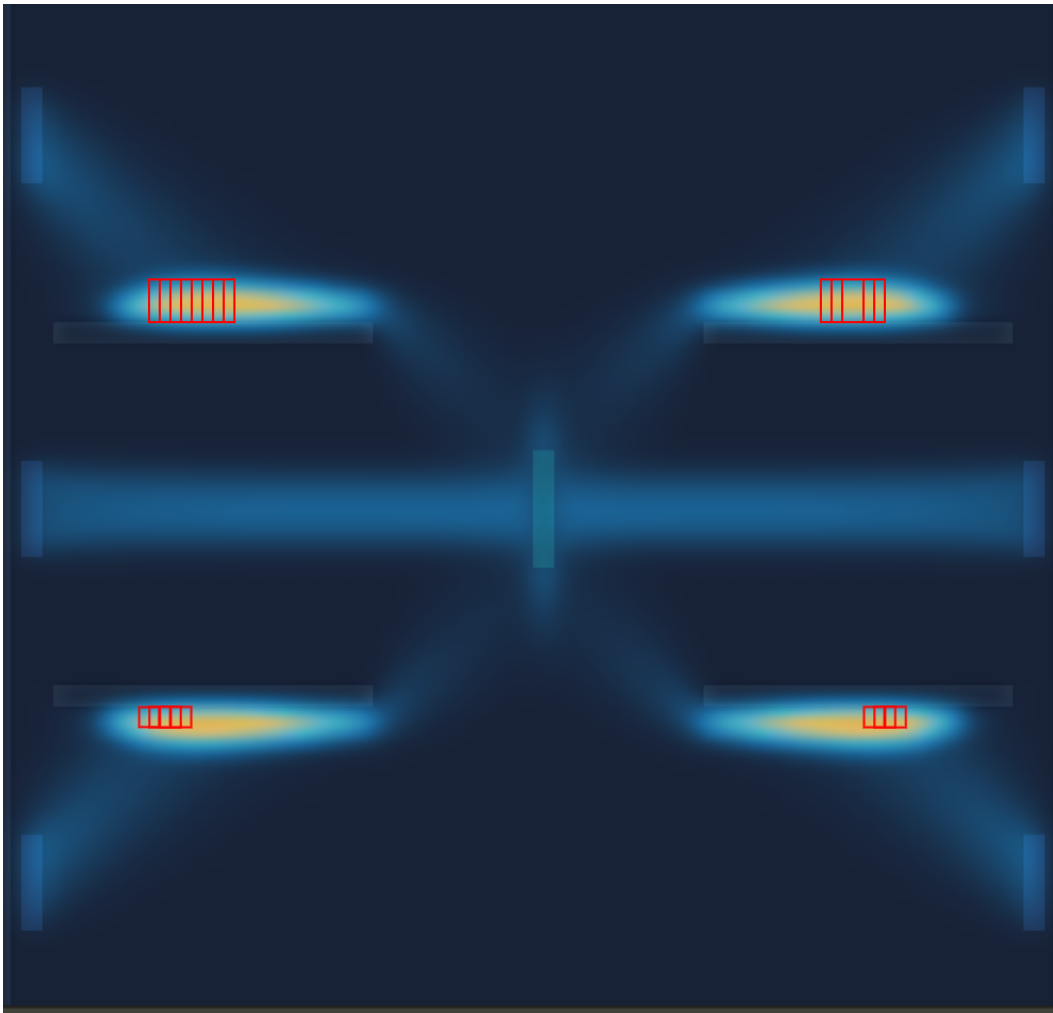


Figure 2: Density evolution during a stadium evacuation scenario. Significant congestion develops upstream of the exit corridor, producing spontaneous lane-like structures and elevated risk levels near the bottleneck.

As crowd density increases, the nonlinear pressure term becomes active and significantly alters the flow structure. Rather than moving uniformly toward the exit, pedestrians redistribute into preferred pathways of lower resistance.

The model predicts:

- Density accumulation upstream of the bottleneck.
- Formation of lane-like structures.
- Flow channelization near the constriction.
- Reduced velocity in highly congested regions.

- Increased risk scores near the exit boundary.

These behaviors are qualitatively consistent with observations reported in pedestrian dynamics literature.

6.3 Quantitative Metrics

To compare different scenarios, several macroscopic indicators were recorded throughout the simulations.

Scenario	Peak Density	Peak Risk	High-Risk Area	Average Velocity	Evacuation Time
Festival Bottleneck	7.50	1.000	6.35%	0.932	400.0
Stadium Rush	6.37	0.961	5.81%	1.014	356.2

Table 2: Representative simulation metrics generated by the proposed model.

The results indicate a strong correlation between increasing density and elevated risk values. Scenarios exhibiting greater congestion produce larger high-risk regions and lower average pedestrian velocities. Regions exceeding the critical density threshold correspond closely with locations identified by the proposed risk functional as potential instability zones.

7 Model Validation

Although the present work does not include direct calibration against experimental datasets, the simulations reproduce several qualitative phenomena commonly reported in empirical and theoretical pedestrian-dynamics studies.

These include:

- Density accumulation upstream of bottlenecks.
- Congestion-induced velocity reduction.
- Lane formation and flow channelization.
- Spatial clustering of high-density regions.

The spontaneous emergence of lane-like structures is particularly noteworthy because no explicit behavioral rules were introduced to generate them. Instead, these patterns arise naturally from interactions between density-dependent mobility, nonlinear pressure effects, and environmental geometry.

The ability of the model to reproduce these known macroscopic phenomena provides preliminary evidence supporting the physical plausibility of the proposed framework.

8 Discussion

The proposed model combines harmonic route guidance, density-dependent mobility, and nonlinear congestion pressure into a unified continuum framework for pedestrian dynamics.

Unlike microscopic agent-based simulations, which become computationally expensive for large populations, the present formulation scales efficiently while preserving key collective behaviors.

One particularly important result is the spontaneous formation of lane-like flow structures near bottlenecks. Such structures emerge naturally from the governing equations and suggest that many crowd instabilities can be explained through interactions between density gradients and geometric constraints rather than detailed individual decision-making.

The risk functional further provides a mechanism for identifying potentially dangerous regions before extreme congestion develops. This capability may prove useful for crowd monitoring, evacuation planning, and real-time safety management.

9 Limitations

Several limitations should be acknowledged:

- The continuum assumption breaks down at very low densities.
- Parameter calibration is required for quantitative prediction.
- The model does not explicitly represent individual decision-making.
- Behavioral heterogeneity is not yet included.

These limitations do not invalidate the framework, but they indicate the range of conditions under which the model is most appropriate.

10 Future Work

Several extensions of the present framework remain possible.

Future research should focus on calibration against empirical pedestrian datasets, allowing quantitative validation of model predictions. Additional developments may incorporate heterogeneous pedestrian populations, adaptive route selection, behavioral decision-making, and dynamic obstacle avoidance.

Integration with live sensor networks and computer-vision-based crowd monitoring systems could enable real-time estimation of crowd density and stampede risk during large public events.

Machine-learning-assisted parameter estimation also represents a promising direction for improving predictive accuracy across different environments and crowd compositions.

11 Conclusion

This work introduces a nonlinear degenerate advection-diffusion framework for modeling crowd motion and predicting stampede risk.

The model combines harmonic route guidance, density-dependent mobility, nonlinear congestion pressure, and a physically motivated risk functional within a unified continuum formulation. Numerical experiments demonstrate the emergence of bottleneck congestion, density accumulation, flow channelization, and lane-like structures commonly observed in pedestrian systems.

The framework remains computationally efficient while reproducing several qualitative characteristics of real crowd dynamics. These properties suggest potential applications in evacuation planning, crowd monitoring, infrastructure design, and real-time risk assessment.

Future validation against experimental and observational datasets will be essential for establishing predictive accuracy and supporting operational deployment.

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